CSCA Chemistry Exam Answers and Detailed Analysis for January 2026

explain

Each question has four options, and only one is correct.
Atomic weights: O: 16, Na: 23, Cu: 64, H: 1, C: 12, He: 4

Question 1

answer： A

Analysis: Chemical properties refer to the characteristics exhibited by substances during chemical reactions, while physical properties manifest without such reactions. Option A (iron nails rusting) demonstrates chemical properties through reactions with oxygen and water. Options B (density), C (turning red when heated − merely a state change), and D (color and state) are all physical properties that can be observed without chemical changes.

Question 2

answer： A

Analysis: Option A: A measuring cylinder is a specialized instrument for precise liquid volume measurement with accurate graduations. Option B: A beaker is primarily used for dissolution and reaction vessels, offering low precision for volume measurement and thus inaccurate results. Option C: The term 'balance' is incorrect; the correct term is 'balance scale,' which measures mass, not volume. Option D: A gas collection bottle is designed for gas gathering and storage, not for liquid volume measurement.

Question 3

answer： D

Analysis: Option A: Iron reacts with concentrated sulfuric acid at room temperature through passivation, halting the reaction and preventing hydrogen production. Option B: Concentrated nitric acid, being highly oxidative, reacts with magnesium to form nitrogen oxides instead of hydrogen. Option C: Copper, being less reactive than hydrogen in the metal activity series, does not react with dilute sulfuric acid. Option D: Zinc reacts with dilute sulfuric acid to produce zinc sulfate and hydrogen ($Zn + H_2SO_4 = ZnSO_4 + H_2\uparrow$), a standard method for hydrogen production in laboratories.

Question 4

answer： C

Analysis: The structural formula of acetic acid is CH_3COOH, containing a carboxyl (-COOH) functional group. Option A (CH_3CH_2OH) represents ethanol's structural formula; option B (CH_3OCH_3) is methyl ether's structural formula; and option D (CH_3CHO) is acetaldehyde's structural formula. The question stem's option C, 'CH3 carboxyl group,' is incorrect. The correct structural formula is CH_3COOH, making option C the correct answer.

Question 5

answer： C

Analysis: ^{13}C and ^{12}C are isotopes (D is correct), and isotopes have the same number of protons and extranuclear electrons (A and B are correct), but different neutron numbers. The neutron number of ^{13}C is 13-6=7, while that of ^{12}C is 12-6=6, so option C is incorrect.

Question 6

answer： A

Analysis: In a solution with pH=1, c(H+) = 0.1 mol·L-¹. Option A: Hydrochloric acid is a strong acid and fully ionizes: $HCl = H^+ + Cl^-$, so c(H+) in 0.1 mol·L-¹ hydrochloric acid is 0.1 mol·L-¹, pH=1; Option B: Sulfuric acid is a binary strong acid and fully ionizes: $H_2SO_4 = 2H^+ + SO_4^{2-}$, so c(H+) in 0.1 mol·L-¹ sulfuric acid is 0.2 mol·L-¹, pH<1; Option C: Sodium hydroxide is a strong base and ionizes to OH^-, making the solution alkaline, pH>7; Option D: Acetic acid is a weak acid and partially ionizes, so c(H+) in 0.1 mol·L-¹ acetic acid is <0.1 mol·L-¹, pH>1.

Question 7

answer： A

Analysis: While potassium produces a purple flame color, the experiment often contains sodium impurities that emit a yellow flame, which obscures the purple color of potassium. The blue cobalt glass serves to filter out the yellow light from sodium, making the purple color of potassium clearly visible (Option A is correct). Option B states that protecting the eyes is not its primary function. Option C clarifies that cobalt glass does not enhance purple light but only filters stray light. Option D confirms that cobalt glass does not affect the observation of temperature.

Question 8

answer： B

Explanation: A solution is a homogeneous and stable mixture formed by dispersing one or several substances into another. Option A: Ethyl acetate is insoluble in water, resulting in layering and forming an emulsion. Option B: Ethanol is miscible with water, forming a homogeneous and stable solution. Option C: Benzene is insoluble in water, leading to layering and forming an emulsion. Option D: Calcium carbonate is a poorly soluble salt, insoluble in water, forming a suspension.

Question 9

answer： C

Analysis: A pure substance consists of a single component. Option A: Medical alcohol is a 75% ethanol-water solution, containing ethanol and water, and is classified as a mixture. Option B: Concrete contains multiple components such as cement, sand, and gravel, and is also classified as a mixture. Option C: Copper is a pure element composed of copper, and is classified as a pure substance. Option D: Normal saline is a 0.9% sodium chloride solution, containing sodium chloride and water, and is classified as a mixture.

Question 10

answer： C

Analysis: Isomers are organic compounds with similar structures, differing in molecular composition by one or several "CH2" atomic groups, and sharing the same types and numbers of functional groups. The unsaturation degree of CgH16 is Ω = (2×9+ 2-16)/2 = 2, classifying it as an alkyne or diene. Option A, C4H8, has an unsaturation degree of 1.

Option ω= (2×4+2-8)/2=1, which may be an olefin or cycloalkane with dissimilar structures; Option B C4H10 has an unsaturation degree ω=0, classified as an alkane with dissimilar structures; Option C C5H8 has an unsaturation degree ω=(2×5+2-8)/2=2, structurally similar to CgH16 but differing by 4 CH2 groups in molecular composition, thus belonging to homologues; Option D C8H10 has an unsaturation degree ω=(2×8+2-10)/2=4, potentially an aromatic hydrocarbon with dissimilar structures.

Question 11

answer： C

Analysis: The oxidizing agent undergoes reduction in the reaction. Option A: The reaction 2NaCl + H2SO4 (concentrated) ▵Na2SO4 + 2HCl↑ is a double decomposition reaction, where the oxidation states of all elements remain unchanged. Sulfuric acid merely provides H+ as an acid. Option B: The reaction H2SO4 + Ba(NO3)2 = 2HNO3 + BaSO 4↓ is a double decomposition reaction, with no change in oxidation states. Sulfuric acid provides SO4 ²-as an acid. Option C: In the reaction S + 2H2SO4 (concentrated) ▵3SO2↑ + 2H2O, the oxidation state of sulfur in sulfuric acid decreases from +6 to +4, undergoing reduction. Sulfuric acid acts as the oxidizing agent. Option D: The reaction 2KOH + H2SO4 = K2SO4 + 2H2O is a neutralization reaction, with no change in oxidation states. Sulfuric acid provides H+ as an acid.

Question 12

answer： D

Analysis: Electrolytes are compounds that conduct electricity in aqueous solutions or molten states. Option A: Sucrose is non-conductive in both aqueous and molten states, thus classified as a non-electrolyte. Option B: Carbon dioxide conducts electricity when dissolved in water because it forms carbonic acid, which is an electrolyte, whereas carbon dioxide itself is non-conductive and thus classified as a non-electrolyte. Option C: Aluminum is a pure element, neither electrolyte nor non-electrolyte. Option D: Sodium chloride conducts electricity in both aqueous solution ($NaCl = Na^+ + Cl^-$) and molten state, thus classified as an electrolyte.

Question 13

answer： D

Analysis: The chemical equation for hydrogen combustion is 2H2 + O2 → 2H2O. The equation shows that 2 mol of H2 completely burns to form 2 mol of H2O, meaning 1 mol of H2 fully burns to produce 1 mol of H2O. Given that the molar mass of water (H2O) is 18 g·mol-¹, the mass of water produced (m) equals n × M = 1 mol × 18 g·mol-¹= 18 g.

Question 14

answer： B

Analysis: Electrolytes are compounds that conduct electricity in aqueous solutions or molten states. Option A: Magnesium is a pure element, neither an electrolyte nor a non-electrolyte. Option B: Sodium chloride conducts electricity in both aqueous solutions and molten states, thus classified as an electrolyte. Option C: Toluene is non-conductive in both aqueous solutions and molten states, therefore categorized as a non-electrolyte. Option D: Helium is a pure element, neither an electrolyte nor a non-electrolyte.

Question 15

answer： A

Analysis: Atoms with stronger metallic properties are more prone to lose electrons and form cations. Option A: Sodium (Na) belongs to Group ⅠA with one outermost electron, exhibiting strong metallic characteristics and readily losing electrons. Option B: Magnesium (Mg) is in Group ⅡA with two outermost electrons, demonstrating weaker metallic properties than Na and less electron-donating capacity. Option C: Chlorine (Cl) in Group ⅦA has seven outermost electrons, readily accepting electrons to form anions. Option D: Fluorine (F) in Group ⅦA possesses seven outermost electrons, exhibiting the strongest non-metallic properties and readily accepting electrons to form anions.

Question 16

answer： D

Analysis: Isomorphs of the same element refer to different pure substances composed of the same element. Option A: CH4 and C3H8 are both hydrocarbons and belong to the same series, not isomorphs. Option B: ^{12}C and ^{14}C are different isotopes of carbon, not pure substances. Option C: CO and CO2 are different oxides of carbon and belong to compounds, not isomorphs. Option D: Diamond and graphite are both pure substances composed of carbon with different structures, belonging to allotropes.

Question 17

answer： A

Analysis: The valence of chlorine is determined by the algebraic sum of the valence states of its constituent elements. Option A: In NaClO3, Na has a +1 valence state and O has a-2 valence state. Let Cl's valence be x. The equation (+1) + x + 3 × (-2) = 0 yields x = +5. Option B: In NaClO4, Na has a +1 valence state and O has a-2 valence state. Let Cl's valence be x. The equation (+1) + x + 4 × (-2) = 0 yields x = +7. Option C: In CaCl2, Ca has a +2 valence state. Let Cl's valence be x. The equation (+2) + 2x = 0 yields x = -1. Option D: In sodium hypochlorite (NaClO), Na has a +1 valence state and O has a-2 valence state. Let Cl's valence be x. The equation (+1) + x + (-2) = 0 yields x = +1.

Question 18

answer： B

Analysis: The atomic number of argon is 18, and its electron configuration follows the "2, 8, 8" rule (2 electrons in the K shell, 8 in the L shell, and 8 in the M shell). Option A, with an atomic number of 9 and electron configuration 2, 7, is fluorine; Option B, with an atomic number of 18 and electron configuration 2, 8, 8, is argon; Option C, with an atomic number of 10 and electron configuration 2, 8, is neon; Option D, with an atomic number of 17 and electron configuration 2, 8, 7, is chlorine.

Question 19

answer： C

Analysis: The downward air displacement method is applicable to gases with a density lower than air and non-reactive with air; the displacement method is suitable for gases insoluble in water and non-reactive with water. Option A, CO2, has a density greater than air (relative molecular mass 44> 29), thus cannot be collected using the downward air displacement method; it is soluble in water and reacts with water, making the displacement method unsuitable. Option B, N2, has a density close to air (relative molecular mass 28 ≈ 29), thus cannot be collected using the air displacement method; it is insoluble in water, making the displacement method applicable. Option C, H2, has a density lower than air (relative molecular mass 2 <29), thus can be collected using the downward air displacement method; it is insoluble in water, making the displacement method applicable. Option D, NH3, has a density lower than air, making the downward air displacement method applicable; it is highly soluble in water, making the displacement method unsuitable.

Question 20

answer： B

Analysis: Solution is a homogeneous and stable mixture whose molar concentration is independent of the volume of the solution. Take 50mL from 1L 1mol·L-' NaOH solution, the homogeneity of the solution remains unchanged, and the molar concentration is still 1mol·L-'.

Question 21

answer： A

Analysis: Substances capable of oxidizing Fe^{2+} to Fe^{3+} must possess oxidizing properties (i.e., the ability to accept electrons). Option A: Cl_2 exhibits strong oxidizing properties and can undergo the reaction: $Cl_2 + 2Fe^{2+} \rightarrow 2Fe^{3+} + 2Cl^-$, which oxidizes Fe^{2+} to Fe^{3+}. Option B: Fe is a single element with reducing properties and cannot oxidize Fe^{2+}. Option C: Zn has reducing properties and can undergo a substitution reaction with Fe^{2+} ($Zn + Fe^{2+} \rightarrow Zn^{2+} + Fe$), but it cannot oxidize Fe^{2+}. Option D: NaOH reacts with Fe^{2+} to form $Fe(OH)_2$ precipitate, which does not involve redox reactions.

Question 22

answer： A

Analysis: Option A, NaHCO3 reacts with hydrochloric acid: NaHCO3 + HCl = NaCl + H2O + CO2↑; reacts with NaOH: NaHCO3 + NaOH = Na2CO3 + H2O, demonstrating its ability to react with both acids and bases; Option B, Ca(OH)2 is a base, reacts with hydrochloric acid (Ca(OH)2 + 2HCl = CaCl2 + 2H2O), but does not react with NaOH; Option C, CH3COOH is an acid, reacts with NaOH (CH3COOH + NaOH = CH3COONa + H2O), but does not react with hydrochloric acid; Option D, NaCl is a salt, does not react with either hydrochloric acid or NaOH.

Question 23

answer： D

Analysis: The variation pattern of atomic radius follows these rules: For elements within the same period, atomic radius decreases progressively from left to right; for elements in the same main group, atomic radius increases gradually from top to bottom. Oxygen (O), carbon (C), and fluorine (F) belong to the second period, while sodium (Na) is in the third period (where third-period elements have larger atomic radii than second-period elements). In the second period, the atomic number of carbon (C) is less than that of fluorine (F), and the atomic radius of carbon (C) is greater than that of fluorine (F). Therefore, fluorine (F) has the smallest atomic radius.

Question 24

answer： A

Analysis: The essence of chemical equilibrium is that the forward reaction rate equals the reverse reaction rate (v forward = v reverse), indicating the reaction system is in dynamic equilibrium (A is correct). Option B states that the concentrations of reactants and products at equilibrium may not be equal, depending on the initial concentrations and equilibrium constants. Option C indicates that the time atoms spend in reactants and products is not necessarily equal. Option D notes that the masses of reactants and products at equilibrium no longer change, though they may not be equal.

Question 25

answer： D

Analysis: The chemical equation for the reduction of cuprous oxide (Cu2O) by H2 is: Cu2O + H2 △2Cu + H2O. The molar quantity n(Cu2O) of 4.0g Cu2O = m/M = 4.0g/(144g·mol⁻¹) = 1/36 mol. From the equation, 1 mol of Cu2O consumes 1 mol of H2, so the molar quantity n(H2) consumed = 1/36 mol. Under standard conditions, the volume V of H2 = n × Vm = 1/36 mol × 22.4 L·mol⁻¹ ≈ 0.622 L.Note:The question may contain an error. If the substance is cupric oxide (CuO), the reaction would be CuO + H2 △Cu + H2O. For 4.0g CuO, the molar quantity n = 4.0g/(80g·mol⁻¹) = 0.05 mol, consuming 0.05 mol of H2. The volume V = 0.05 mol × 22.4 L·mol⁻¹ = 1.12 L (corresponding to option D). Based on the options, the term "cuprous oxide" in the question should be "cupric oxide," making D the correct answer.

Question 26

answer： C

Analysis: In NaNO2, Na stands for sodium (sodium), and NO2-denotes nitrite (nitrite), hence the English name is sodium nitrite. Option A, Ammonium nitrate is NH4NO3 (ammonium nitrate); Option B, Sodium nitride is Na3N (sodium nitride); Option D, Sodium nitrate is NaNO3 (sodium nitrate).

Question 27

answer： D

Analysis: The higher the solution pH, the stronger the alkalinity. Option A: Sodium chloride is neutral in water (pH=7). Option B: HNO3 is a strong acid and acidic in water (pH<7). Option C: H2S is a weak acid and acidic in water (pH<7). Option D: NH3 dissolves in water to form NH3·H2O, which is a weak base that ionizes to OH- (NH3·H2O = NH4++OH-), resulting in an alkaline solution (pH>7). Therefore, option D has the highest pH.

Question 28

answer： D

Analysis: Option A, a decomposition reaction is a chemical reaction in which one substance is divided into two or more substances (e.g., 2KMnO4 △K2MnO4+MnO2+O2 ↑); Option B, a displacement reaction is a chemical reaction in which one element and one compound react to form another element and another compound (e.g., Fe + CuSO4 = FeSO4 + Cu); Option C, a combination reaction is a chemical reaction in which two or more substances combine to form one substance (e.g., 2H2 + O2 → 2H2O); Option D, a neutralization reaction is a chemical reaction in which an acid reacts with a base to form a salt and water. For example, hydrochloric acid (acid) reacts with sodium hydroxide (base): HCl + NaOH = NaCl + H2O, which is a neutralization reaction.

Question 29

answer： A

Analysis: Ionic bonds exist in ionic compounds (containing both anions and cations), while covalent bonds occur in covalent compounds or atomic groups within ionic compounds. Option A: Na2O2 is an ionic compound composed of Na+ and O2 ²- (peroxynitrite ions). The Na+ and O2 ²-ions form an ionic bond, while the two O atoms in O2 ²-share a covalent bond, thus containing both ionic and covalent bonds. Option B: CO2 is a covalent compound containing only the C=O covalent bond. Option C: H2O is a covalent compound containing only the H-O covalent bond. Option D: Potassium chloride (KCl) is an ionic compound.

The compound contains only ionic bonds between K+ and Cl-.

Question 30

answer： B

Analysis: A substitution reaction is a chemical reaction where one element and one compound react to form another element and another compound. Option A, $2NaHCO_3 \xrightarrow{\triangle} Na_2CO_3 + H_2O + CO_2\uparrow$, represents a decomposition reaction where one substance is converted into three substances. Option B, $Fe + CuSO_4 = FeSO_4 + Cu$, is a substitution reaction where Fe (element) reacts with $CuSO_4$ (compound) to form $FeSO_4$ (compound) and Cu (element). Option C, $2H_2 + O_2 \rightarrow 2H_2O$, is a combination reaction where two elements combine to form one compound. Option D, $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$, is an oxidation-reduction reaction (combustion reaction) where a compound reacts with an element to form two compounds, and thus does not qualify as a substitution reaction.

Question 31

answer： A

Analysis: The reaction for producing oxygen using hydrogen peroxide (H_2O_2) in the laboratory is: $2H_2O_2 \rightarrow MnO_2 + O_2\uparrow$, where MnO_2 (manganese dioxide) acts as a catalyst to accelerate the reaction rate. Options B (potassium chloride, KCl), C (sodium hydroxide, NaOH), and D (potassium chlorate, $KClO_3$) cannot catalyze the decomposition of H_2O_2.

Question 32

answer： D

Analysis: Option A: $KClO_3$ itself lacks bleaching properties. It only exhibits bleaching activity under acidic conditions or through thermal decomposition to produce ClO_2. Option B: Dry Cl_2 is non-bleaching. When Cl_2 reacts with water, it forms HClO ($Cl_2 + H_2O = HCl + HClO$), which possesses bleaching properties. Option C: Sodium chloride (NaCl) is non-bleaching. Option D: HClO (hypochlorous acid) demonstrates strong oxidizing properties, capable of disrupting the structure of colored substances, and thus exhibits bleaching activity (the bleaching principles of products like bleach powder and 84 disinfectant both involve the formation of HClO).

Question 33

answer： C

Analysis: In the periodic table, the period number equals the number of electron shells, and the main group number equals the number of electrons in the outermost shell. The electron number of elements in the second period is 2. The outermost electron count of Group VIIA elements is 7. Option A: Oxygen (O) belongs to Group VIIA in the second period; Option B: Nitrogen (N) belongs to Group VIA in the second period; Option C: Fluorine (F) belongs to Group VIIIA in the second period; Option D: Chlorine (Cl) belongs to Group VIIIA in the third period.

Question 34

answer： A

Analysis: Conditions for coexistence of large amounts of ions: No reactions occur between the ions (no precipitation, gas, water, or weak electrolyte formation, and no oxidation).

Reduction reaction). Option A: Mg^{2+}, Cl^-, NH_4^+, and SO_4^{2-} do not react and can coexist in large quantities; Option B: Ca^{2+} reacts with SO_4^{2-} to form the sparingly soluble $CaSO_4$, while Ca^{2+} reacts with CO_3^{2-} to form $CaCO_3$ precipitate, so they cannot coexist in large quantities; Option C: CO_3^{2-} reacts with H^+ to form H_2O and CO_2 ($CO_3^{2-} + 2H^+ \rightarrow H_2O + CO_2\uparrow$), so they cannot coexist in large quantities; Option D: OH^- reacts with Ca^{2+} to form the sparingly soluble $Ca(OH)_2$, while OH^- reacts with Mg^{2+} to form $Mg(OH)_2$ precipitate, so they cannot coexist in large quantities.

Question 35

answer： A

Analysis: The solute mass fraction = (solute mass/solution mass) × 100%. Option A: When a portion of solvent is evaporated at constant temperature, the saturated solution remains saturated, and the solute mass fraction remains unchanged. This statement may contain a wording error in the question stem. If it's an unsaturated solution, evaporating solvent at constant temperature would reduce the total solution mass, increasing the mass fraction. If it's a saturated solution, evaporating solvent would cause solute precipitation, leaving the solution saturated and the mass fraction unchanged. Combining the options, the most reasonable answer is A (presuming "saturated" in the question stem is a typographical error or considering the solution remains unsaturated after solvent evaporation). Option B: The solubility of potassium nitrate increases with temperature. Raising the temperature of the saturated solution maintains both solute and solvent masses, keeping the mass fraction unchanged. Option C: Lowering the temperature of the saturated solution causes potassium nitrate to precipitate, reducing the solute mass while solvent mass remains unchanged, resulting in a decreased mass fraction. However, the question doesn't specify whether "saturated" refers to low-temperature conditions, and Option A directly alters the mass fraction. Option D: Adding solid potassium nitrate to a saturated potassium nitrate solution doesn't dissolve the solid, maintaining both solute and solvent masses, and thus the mass fraction remains unchanged.

Question 36

answer： D

Analysis: Method for detecting Cl^-: Add $AgNO_3$ solution to the solution to form a white precipitate, then add dilute nitric acid. The precipitation does not dissolve (eliminating interference from impurity ions such as CO_3^{2-}). Option A: $Ba(NO_3)_2$ does not react with Cl^-, making it unsuitable for detection. Option B: $AgNO_3$ reacts with Cl^- to form AgCl precipitate, but SO_4^{2-} in dilute sulfuric acid forms a slightly soluble Ag_2SO_4 compound with Ag^+, interfering with detection. Option C: $BaCl_2$ does not react with Cl^-, making it unsuitable for detection. Option D: $AgNO_3$ reacts with Cl^- to form AgCl precipitate, and dilute nitric acid can eliminate impurity ions (e.g., CO_3^{2-}).
2026 January CSCA Chemistry Exam Answers and Detailed Analysis

Question 37

answer： A

Analysis: Calculate the number of atoms for each option: Option A, standard condition: 11.2L CH 4's molar quantity n=V/Vm=11.2L/22.4L·mol^{-1}=0.5mol, 1 CH4 molecule contains 5 atoms, atomic number=0.5mol×5×Na=2.5Na; Option B, 2g He's molar quantity n=2g/4g·mol^{-1}=0.5mol, He is a monatomic molecule, atomic number=0.5mol×Na=0.5Na; Option C, 3.01×10^{23} Al atoms = 3.01×10^{23}/Na≈0.5Na; Option D, 1mol HF contains 2 atoms (H and F), atomic number=1 mol×2×Na=2Na. Therefore, the option with the most atomic number is Option A.

Question 38

answer： C

Analysis: Le Chatelier's principle states that altering conditions affecting equilibrium shifts the reaction toward the direction that weakens the change. The reaction C(s) + H2O(g) = CO(g) + H2(g) has a ΔH> 0 (endothermic), with an increased number of gas molecules on the right side (2 mol) compared to the left side (1 mol). Option A: Increasing temperature (endothermic reaction, equilibrium shifts forward) and pressure (equilibrium shifts toward fewer gas molecules, i.e., backward) makes the overall direction of equilibrium uncertain. Option B: Decreasing temperature (equilibrium shifts backward) and pressure (equilibrium shifts forward) also makes the overall direction uncertain. Option C: Decreasing temperature (equilibrium shifts backward) and increasing pressure (equilibrium shifts backward) shifts the equilibrium toward the reverse reaction. Option D: Increasing temperature (forward) and decreasing pressure (forward) shifts the equilibrium forward.

Question 39

answer： D

Analysis: The purification of H2 requires the removal of CO2 and NH3, with no introduction of new impurities. NH 3 is an alkaline gas, while CO2 is an acidic gas. The first step involves removing CO2 and NH3: NaOH solution can absorb CO2 (2NaOH + CO2 = Na2CO3 + H2O) and NH3 (NH3 + H2O = NH3·H2O). The second step requires drying H2 (removing water vapor carried by the NaOH solution). Concentrated sulfuric acid, which has hygroscopic properties, can dry H2. Option A: First adding concentrated sulfuric acid cannot absorb CO2, followed by NaOH solution absorbing CO2 but introducing water vapor, leaving H2 un-dried. Option B: Calcium oxide absorbs water vapor and NH3, while NaOH solution absorbs CO2, but ultimately H2 remains un-dried. Option C: NaOH solution absorbs both CO2 and NH3, and calcium oxide absorbs water vapor, but the reaction between calcium oxide and water produces Ca(OH)2, potentially introducing impurities. Option D: NaOH solution absorbs both CO2 and NH3, and concentrated sulfuric acid dries H2, achieving optimal purification.

Question 40

answer： A

Analysis: The combustion reaction of carbon: ①C + O2 produces CO2 (when O2 is sufficient); ②2C + O2 produces 2CO (when O2 is insufficient). The molar quantity n(C) of 12g carbon = 12g/12g·mol^{-1} = 1mol, reacting with 0.8mol O2. If CO2 is fully formed, 1mol of O2 is required (as shown in ①), but only 0.8mol of O2 is available, resulting in partial formation of CO2 and CO. Therefore, the generated gas may contain both CO and CO2 (Option A is correct). Option B: If only CO is formed, 0.5mol of O2 is needed (as shown in ②), but 0.8mol of O2 is available, indicating excess O2, so CO alone cannot be formed. Option C: Producing both CO2 and O2 requires more than 1mol of O2, but only 0.8mol is available, making this impossible. Option D: Producing only CO2 requires 1mol of O2, which is insufficient, making this impossible.

Question 41

answer： C

Analysis: NaHCO3 decomposes readily upon heating: 2NaHCO3 $\triangle$Na2CO3 + H2O + CO2 ↑, whereas Na 2CO3 remains stable under heat. Thus, heating effectively removes NaHCO3 from Na2CO3 (Option C is correct). Option A introduces CO2, causing Na2CO3 to react with CO2 and H2O to form NaHCO3 (Na2 CO3 + CO2 + H2O = 2NaHCO3), which increases impurities. Option B adds NaOH, where NaHCO3 reacts with NaOH to form Na2CO3 (NaHCO3 + NaOH = Na2CO3 + H2O), but introduces new impurity NaOH (excess). Option D adds dilute hydrochloric acid, reacting both Na2CO3 and NaHCO3 (Na2CO3 + 2HCl = 2NaCl + H2O + CO2 ↑, NaHCO3 + HCl = NaCl + H2O + CO2 ↑), depleting the primary substances.

Question 42

answer： D

Analysis: The reaction 2HI(g) = H2(g) + I2(g) ΔH<0 (exothermic) maintains constant gas molecular numbers (2 mol on both sides). Option A: Increasing temperature reverses the exothermic equilibrium. Option B: Catalysts only accelerate the reaction rate without affecting equilibrium. Option C: Increasing system pressure maintains constant gas molecular numbers, keeping equilibrium unchanged. Option D: Removing H2 (reducing product concentration) shifts equilibrium toward the forward reaction.

Question 43

answer： C

Analysis: This reaction is an esterification between acetic acid and ethanol (D is correct). Concentrated sulfuric acid serves as both a catalyst (accelerating the reaction rate) and an absorbent (absorbing the generated water, shifting the equilibrium forward, and improving yield) (A and B are correct). Esterification is a reversible reaction, but reversible reactions cannot proceed to completion, as reactants cannot be fully converted into products (C is incorrect).

Question 44

answer： B

Analysis: The chemical equation for the reaction of Na with water is: 2Na + 2H2O=2NaOH + H2↑. When 0.1 mol of Na reacts, it consumes 0.1 mol of H2O (mass m=0.1 mol×18 g·mol⁻¹=1.8 g) and produces 0.1 mol of NaOH (mass m=0.1 mol×40 g·mol⁻¹=4 g) and 0.05 mol of H2 (mass m=0.05 mol×2 g·mol⁻¹=0.1 g). The mass of the solution after the reaction is 37.8 g (mass of water) + 2.3 g (mass of Na).
-0.1g (mass of H2) = 40g (mass of Na = 0.1mol × 23g·mol-'= 2.3g). The solute mass fraction = (4g/40g) × 100% = 10%.

Question 45

answer： B

Analysis: The nuclear charge number of the ground state potassium atom is 19, and the electron configuration follows the principle of minimum energy, with the electron configuration being 1s ² 2s ² 2p⁶ 3s ² 3p⁶ 4s '(the 4s orbital energy is lower than the 3d orbital, so electrons are first filled in the 4s orbital). Option A, the 2p orbital can hold up to 6 electrons, and the $2p^7$ configuration is incorrect; Option C, the electron configuration is repetitive (3s ² appears twice), and the 4p orbital energy is higher than the 4s orbital, so the configuration is incorrect; Option D, the 2s ' and $2p^4$ configurations do not conform to the principle of minimum energy, so it is incorrect.

Question 46

answer： D

Analysis: Based on experimental observations: ① Adding sufficient water causes partial dissolution of solids; ② Adding sufficient dilute hydrochloric acid produces bubbles (containing CO3 ²-ions) and complete dissolution of solids; ③ Adding sufficient dilute sulfuric acid generates bubbles, but residual solids remain after shaking (forming acid-insoluble precipitates). Option A: Both CaCO3 and BaCO3 are insoluble in water (discrepancy with ①), while dilute hydrochloric acid causes dissolution with bubble formation. Dilute sulfuric acid produces CaSO4 and BaSO4 precipitates, but solids remain insoluble in ①, thus excluded. Option B: AgNO3 and Na2CO3 are soluble in water (discrepancy with ①), and dilute hydrochloric acid forms AgCl precipitate (incomplete dissolution of solids, discrepant with ②), thus excluded. Option C: NaOH and CuSO4 are soluble in water, forming Cu(OH)2 precipitate (partial dissolution of solids in ①). Dilute hydrochloric acid dissolves the precipitate (no bubbles, discrepant with ②), thus excluded. Option D: Na2CO3 is soluble in water while BaCO3 is insoluble (consistent with ①). Dilute hydrochloric acid produces CO2 bubbles and complete dissolution of solids (consistent with ②). Dilute sulfuric acid generates CO2 bubbles for Na2CO3 and BaSO4 precipitates (residual solids after shaking).

③ The phenomenon is consistent with), which is correct.

Question 47

answer： C

Analysis: Deduced elements: The outermost electron count of X is twice that of its innermost shell. In short-period elements, the innermost shell has 2 electrons (second period) and the outermost shell 4 electrons, thus X is C (carbon). The M-shell electron count of Y is half that of the L-shell (L-shell 8 electrons vs. M-shell 4 electrons), making Y Si (silicon). Z's innermost shell has 1 more electrons than its outermost shell. In short-period elements, the innermost shell has 8 electrons (third period) and the outermost 7, thus Z is Cl (chlorine). Option A: The compound formed by X and Z is CCl_4, insoluble in water, correct. Option B: C, Si, and Cl are all non-metallic elements, correct. Option C: X (C) belongs to Group ⅣA, and Y (Si) also belongs to Group ⅣA, same main group, correct? Here, there may be an inference error. If Z is N (innermost shell 2, outermost shell 1, inconsistent) or X is He (outermost shell 2, innermost shell 1, inconsistent), the original inference would be correct, making option C correct. There may be an error in the question stem. Re-inference: Z's innermost shell has 1 more electrons than its outermost shell. If Z is N (second period, innermost shell 2, outermost shell 1, inconsistent); if Z is P (third period, innermost shell 8, outermost shell 5, inconsistent), the original inference that Z is Cl is correct. If the incorrect option is C, or if X is O (outermost layer 6, second outermost layer 2, which is inconsistent), and the inference is correct, then the statement 'same group' in the question could refer to 'same cycle,' making C incorrect. Considering the options, C is the most likely incorrect choice.

Question 48

answer： A

Analysis: The chemical equation for electrolyzing saturated salt water is: $2NaCl + 2H_2O \rightarrow 2NaOH + H_2\uparrow + Cl_2\uparrow$. The cathode (connected to the negative terminal of the power source) produces H_2 and NaOH, while the anode (connected to the positive terminal) generates Cl_2. Option A: If port a is the cathode, the reaction produces NaOH (the solution becomes alkaline), and phenolphthalein turns red after addition, which is correct. Option B: If port a is the cathode, H_2 is produced, but it does not cause the moist litmus paper to change color. If it is the anode, Cl_2 is produced, which first turns the moist litmus paper red and then fades, which is incorrect. Option C: If port b is the anode, Cl_2 is produced, but Cl_2 cannot burn. If it is the cathode, H_2 is produced, and burning produces a pale blue flame, which is incorrect. Option D: In the circuit, for every 1 mol of electrons transferred, 0.5 mol of H_2 or 0.5 mol of Cl_2 is produced. Under standard conditions, the volume is 11.2 L, which is incorrect. Based on the principle of electrolysis, the cathode produces NaOH, so option A is correct.